Across CES

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Abstract

I show that CES share functions have an exact representation as log-logistic distribution functions in own price, and that any demand system nesting CES inherits a generalization of this structure. If the generalized share functions are monotone in own price and admit limits at zero and infinity, they are themselves distribution functions. The result is purely representational: it requires neither utility maximization nor integrability.

Because share functions fully characterize a demand system, identifying them with distribution functions opens the language of distribution theory — monotonicity, tail behavior, hazard rates, and parametric families — as a toolkit for demand system design. The identification also has economic content: each share function is the survival function of a reservation price distribution in an equivalent atomistic economy, where a continuum of unit-demand consumers generate the same aggregate demand through binary purchasing decisions. The representation is constructive: choosing a family of distribution functions generates a demand system that nests CES by construction.

1 Introduction

CES demand systems are ubiquitous in economic modeling. Their appeal is tractability: closed-form demands, a simple price index, transparent comparative statics. Their limitation is equally well known: uniform substitution patterns, no local complementarity, and curvature restrictions that are often at odds with empirical evidence. A large literature proposes alternatives — nested CES, variable-elasticity aggregators, flexible functional forms for indirect utility — but these typically modify primitives (utility functions, price aggregators) and inherit their own structural constraints.

I take a different approach. Any demand system can be written as $x_j = (Y/p_j)\, s_j(\mathbf{p}, Y)$, where the share functions $\mathbf{s}(\cdot)$ satisfy positivity and adding-up. Share functions fully characterize the demand system. The key observation is that, in CES, each share function $s_j(p_j; \mathbf{p}_{-j}, Y)$ — viewed as a function of own price — coincides with the cumulative or complementary cumulative distribution function (CDF or CCDF) of a log-logistic distribution.[1]

This fact extends well beyond CES. I show that any demand system nesting CES — in the sense that CES is recovered for some parameter values or limits — induces share functions that are generalizations of the log-logistic (C)CDF. If these generalized shares are monotone in own price and admit limits

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[1] No stochastic choice or random utility is introduced here; the CDF structure is used as a characterization device. Section 4 shows that the distributional representation also admits a structural interpretation through an equivalent atomistic economy.

at zero and infinity, they are themselves (complementary) distribution functions (Theorem 1). The result requires neither utility maximization nor integrability.

The identification of share functions with distribution functions is not merely classificatory — it is constructive. Choosing a family of distribution functions generates a demand system that nests CES by construction. The entire toolkit of distribution theory — monotonicity, tail behavior, hazard rates, parametric families — becomes available for demand system design. Existing generalizations of CES fit naturally within this framework: the Homothetic Single Aggregator (HSA) and Homothetic Directly Integrable Aggregator (HDIA) classes of **?** can be rewritten as generalized log-logistic share systems. The representation also accommodates localized departures from global substitution patterns, including pairs of goods that are complements while all others remain substitutes. Moreover, the distributional structure has economic content: each share function is the survival function of a reservation price distribution in an equivalent atomistic economy, paralleling the aggregation logic of discrete choice models.

Section 2 introduces the framework and states the main result. Section 3 applies the characterization to existing demand systems and constructs new ones. Section 4 shows that the distributional representation has economic content: each share function is the survival function of a reservation price distribution in an equivalent atomistic economy. The Appendix discusses integrability and hazard rate extensions.

2 Demand Systems

Consider an n-dimensional consumption space with consumption bundles $\mathbf{x} \in \mathbb{R}^n_{++}$, prices $\mathbf{p} \in \mathbb{R}^n_{++}$, and income $Y \in \mathbb{R}_{++}$.

2.1 Share functions

Any demand system $\mathbf{x}(\mathbf{p}, Y)$ can be written as

$$x_j(\mathbf{p}, Y) = \frac{Y}{p_j} s_j(\mathbf{p}, Y), \tag{1}$$

where the *share functions* $s_j(\cdot)$ satisfy

$$s_j(\mathbf{p}, Y) > 0, \qquad \sum_{j=1}^{n} s_j(\mathbf{p}, Y) = 1.$$

Equation (1) is an identity — it imposes no behavioral restrictions. Given a collection of share functions, define the *admissible set*

$$E \equiv \left\{ (\mathbf{p}, Y) \in \mathbb{R}^{n+1}_{++} \ : \ \sum_{j=1}^{n} s_j(\mathbf{p}, Y) = 1 \right\}.$$

For $(\mathbf{p}, Y) \in E$, the budget constraint holds exactly. Whether E coincides with $\mathbb{R}^{n+1}_{++}$ or is a strict subset depends on the structure of the share functions.[2]

[2] In CES, $E = \mathbb{R}^{n+1}_{++}$. In demand systems built around a single price aggregator, E may be a strict subset. The representation results are agnostic about the size of E; when constructing generalized share systems, ensuring that E is nonempty or coincides with the full domain requires additional restrictions on the functional form of the shares.

2.2 CES and the log-logistic structure

Fix j and hold $(\mathbf{p}_{-j}, Y)$ constant. The CES share function is

$$s_j(p_j; \mathbf{p}_{-j}, Y) = \frac{1}{1 + p_j^{\sigma-1} \sum_{k \neq j} \left(\frac{\alpha_k}{\alpha_j}\right)^{\sigma} p_k^{1-\sigma}}, \tag{2}$$

where $\sigma > 0$ and $\alpha_j > 0$.

As a function of p_j, expression (2) maps $\mathbb{R}_{++}$ into $(0, 1)$, is monotone, and admits limits as $p_j \to 0$ and $p_j \to \infty$. It therefore coincides with a cumulative or complementary cumulative distribution function of a log-logistic distribution, depending on whether $\sigma < 1$ or $\sigma > 1$.[3]

2.3 Generalizing CES

A family of demand systems $\{\tilde{\mathbf{x}}(\mathbf{p}, Y; \boldsymbol{\gamma}) : \boldsymbol{\gamma} \in \boldsymbol{\Gamma}\}$ *generalizes CES* if there exists $\tilde{\boldsymbol{\gamma}} \in \overline{\boldsymbol{\Gamma}}$ such that

$$\tilde{\mathbf{x}}(\mathbf{p}, Y, \tilde{\boldsymbol{\gamma}}) = \mathbf{x}^{\text{CES}}(\mathbf{p}, Y) \quad \text{for all } (\mathbf{p}, Y) \in \mathbb{R}_{++}^{n+1}.$$

Equivalently, the associated share functions generalize CES shares if $\tilde{s}_j(\cdot; \tilde{\boldsymbol{\gamma}}) = s_j^{\text{CES}}(\cdot)$ for each j. No further structure is imposed: the generalized system need not preserve homogeneity, separability, or any other property of CES away from $\tilde{\boldsymbol{\gamma}}$.

Theorem 1. *Suppose $\tilde{\mathbf{x}}(\mathbf{p}, Y, \boldsymbol{\gamma})$ generalizes a CES demand system. Then:*

(i) For each good j, the CES share $s_j^{\text{CES}}(p_j; \mathbf{p}_{-j}, Y)$ coincides with a (complementary) cumulative distribution function of a log-logistic distribution in own price p_j.

(ii) For each j, the generalized share $\tilde{s}_j(p_j; \mathbf{p}_{-j}, Y, \boldsymbol{\gamma})$ coincides with a generalization of this log-logistic function.

(iii) If $\tilde{s}_j(\cdot)$ is monotone in p_j and admits limits at zero and infinity, then $\tilde{s}_j(\cdot)$ itself coincides with a (complementary) cumulative distribution function.

The theorem is a representation result with a constructive converse: specifying a family of distribution functions generates a demand system that nests CES by construction.

2.4 Construction

A direct generalization of (2) is

$$\tilde{s}_j(p_j; \mathbf{p}_{-j}, Y) = \frac{1}{1 + \frac{h(p_j)}{p_j} \sum_{k \neq j} \frac{p_k}{h(p_k)}}, \tag{3}$$

where $h : \mathbb{R}_{++} \to \mathbb{R}_{++}$ is strictly increasing. When $h(p) = p^{\sigma}$, expression (3) reduces to CES with symmetric taste parameters. The choice of $h(\cdot)$ governs curvature and substitution patterns.

[3] The CDF/CCDF distinction coincides with the $\sigma < 1$ (complements) and $\sigma > 1$ (substitutes) regimes. The distributional language is representational and does not classify gross substitutes versus gross complements in general.

3 Examples and Applications

I illustrate the representation by rewriting two existing demand system classes in the form characterized by Theorem 1, and then constructing a new demand system with localized complementarity.

3.1 HSA and HDIA as generalized log-logistic systems

HSA. In HSA demand systems (**?**), zero homogeneity is achieved through a linearly homogeneous price aggregator $A(\mathbf{p})$:

$$s_j(\mathbf{p}, Y) = s_j\left(\frac{p_j}{A(\mathbf{p})}\right).$$

As in CES, HSA cannot combine gross substitutes and gross complements, the admissible domain E^{HSA} is compact, and each share admits limits at zero and one in own price. Rewriting HSA shares using the adding-up condition shows that each $s_j(\cdot)$ is a generalized log-logistic (C)CDF.

HDIA. HDIA demand systems allow shares to depend on levels of consumption, not only relative prices. The shares can be written as

$$s_j(\mathbf{p}, Y) = \frac{1}{1 + \frac{1}{p_j x_j\left(\frac{p_j}{B(\mathbf{p})}\right)} \sum_{k \neq j} p_k x_k\left(\frac{p_k}{B(\mathbf{p})}\right)},$$

which is again a generalized log-logistic (C)CDF. Both classes fit within the representation of Theorem 1.

3.2 Localized complementarity

CES and the generalizations above impose global substitution patterns: all goods are either mutual gross substitutes or mutual gross complements. The representation also accommodates localized departures. Consider a demand system in which all goods are gross substitutes except a single pair (j, l), which are gross complements. Modify only the share functions for j and l:

$$\begin{aligned}
s_l(p_l; \mathbf{p}_{-l}, Y) &= \frac{1}{1 + p_l^{\sigma-1} \sum_{k \neq l} \left(\frac{\alpha_k}{\alpha_l}\right)^{\sigma} p_k^{1-\sigma} + \gamma p_j^{\mu}}, \\
s_j(p_j; \mathbf{p}_{-j}, Y) &= \frac{1}{1 + p_j^{\sigma-1} \sum_{k \neq j} \left(\frac{\alpha_k}{\alpha_j}\right)^{\sigma} p_k^{1-\sigma} + f}, \\
s_i(p_i; \mathbf{p}_{-i}, Y) &= \frac{1}{1 + p_i^{\sigma-1} \sum_{k \neq i} \left(\frac{\alpha_k}{\alpha_i}\right)^{\sigma} p_k^{1-\sigma}}, \qquad i \neq j, l,
\end{aligned}$$

where $\mu > \sigma > 1$, $\gamma > 0$, and the adjustment term f is chosen to ensure adding-up. The term γp_j^{μ} in s_l makes good l complementary to good j: an increase in p_j raises s_l. The CES structure for all other goods is preserved. A formal analysis of existence and smoothness of the implicit adjustment term f is left for future work.

4 Demand-Side Equivalence

The distributional structure identified in Theorem 1 admits an economic interpretation. Consider an *atomistic economy* for good j: a continuum of unit-demand consumers, each characterized by a reservation price u, distributed according to a CDF F_j. Consumer u purchases good j if and only if $u \geq p_j$. Aggregate demand is

$$x_j(p_j) = \frac{Y}{p_j}\big[1 - F_j(p_j)\big], \tag{4}$$

where Y/p_j converts the expenditure share into a quantity. Comparing with equation (1):

$$s_j(p_j; \mathbf{p}_{-j}, Y) = 1 - F_j(p_j).$$

The share function is the survival function of the reservation price distribution.

In CES, the reservation prices follow a log-logistic distribution. An economy with heterogeneous CES consumers whose individual demands aggregate to (2) is equivalent — in the sense of generating identical aggregate demands — to an atomistic economy in which F_j is log-logistic with shape parameter $\sigma - 1$ and scale determined by $(\mathbf{p}_{-j}, Y)$.[4] By Theorem 1, any demand system nesting CES corresponds to an atomistic economy with generalized log-logistic reservation price distributions.

The equivalence gives the hazard rate a direct interpretation. Writing $\lambda_j(p_j) = f_j(p_j)/[1 - F_j(p_j)]$, where f_j is the density of F_j:

$$\lambda_j(p_j) = \frac{\text{density of consumers at the margin}}{\text{remaining buyers}}.$$

The hazard rate measures the thinness of demand at the margin — the rate at which buyers are lost per unit price increase, relative to those still purchasing. In CES, $\lambda_j(p_j) = (\sigma - 1)/p_j \cdot (1 - s_j)$. Choosing a different distribution family changes the hazard profile, and therefore the substitution behavior, with transparent economic content: a Weibull survival function produces uniformly accelerating or decelerating buyer attrition, while a Burr XII adds a parameter controlling how quickly the marginal buyer density thins in the tails.

The construction parallels discrete choice models. Both aggregate binary decisions into continuous market outcomes. In discrete choice, random utility shocks generate choice probabilities; here, heterogeneous reservation prices generate expenditure shares. The primitives differ — random utility versus reservation prices — but the aggregation logic is the same. The distributional toolkit of Theorem 1 applies to both.

5 Conclusion

CES share functions are log-logistic distribution functions in own price, and any demand system nesting CES inherits a generalization of this structure. The identification is constructive: choosing a family of distribution functions generates a demand system. The atomistic economy equivalence gives the construction economic content: each distribution describes a population of unit-demand consumers whose binary purchasing decisions aggregate into the demand system.

[4] The equivalence is demand-side only: two different microeconomic environments produce the same aggregate demand function. It does not require specifying supply or equilibrium.

Two extensions follow naturally. First, characterizing conditions under which distribution-based share systems are integrable and admit a utility representation. Second, studying equilibrium existence and uniqueness in models with localized departures from global substitution patterns — the complementarity example in Section 3 suggests both the promise and the open questions of such constructions.

Appendix

A. Integrability

Not every collection of positive, adding-up share functions can be rationalized by a utility function — Slutsky symmetry and negativity are additionally required. The representation in Theorem 1 is silent on integrability: it characterizes the shape of share systems, not whether they arise from utility maximization. That said, CES and several of its generalizations (including HSA) satisfy integrability and fit within the framework. Characterizing which distribution-based share systems are integrable is a natural next step.

B. Hazard rate profiles

Section 4 established that the share function $s_j(p_j)$ is the survival function of a reservation price distribution, with hazard rate $\lambda_j(p_j) = f_j(p_j)/[1 - F_j(p_j)]$ measuring the thinness of demand at the margin. Different distribution families generate different hazard rate profiles, and therefore different substitution patterns. This appendix derives the CES hazard rate in closed form and surveys alternatives.

In CES, writing $B_{-j} \equiv \sum_{k \neq j} (\alpha_k/\alpha_j)^\sigma p_k^{1-\sigma}$, the hazard rate simplifies to

$$\lambda^{\text{CES}}(p_j) = \frac{\sigma - 1}{p_j}\big(1 - s_j(p_j)\big).$$

The shape of this hazard rate depends on σ. For $\sigma > 2$, it is inverted-U: rising for low p_j and falling for high p_j. For $1 < \sigma \leq 2$, it is monotone decreasing. This profile — determined entirely by the log-logistic distributional form — governs how rapidly expenditure share is lost as price increases.

Replacing the log-logistic with a different distribution family changes the hazard profile and thus the substitution behavior. For instance:

- A *Weibull* survival function generates a monotone hazard rate — increasing, decreasing, or constant depending on the shape parameter. This produces share functions in which price sensitivity either accelerates or decelerates uniformly, unlike the non-monotone CES pattern.
- A *log-normal* survival function generates a hazard rate that rises from zero, peaks, and then declines slowly — qualitatively similar to the log-logistic but with heavier tails and different curvature in the intermediate range.
- A *Burr Type XII* distribution nests the log-logistic as a special case and adds a parameter controlling tail thickness. It provides a one-parameter extension of CES that preserves the inverted-U hazard shape while allowing the tails to be heavier or lighter.

The Burr XII nests CES exactly; the Weibull and log-normal approximate it in appropriate limits. In each case, the departure from CES substitution patterns is transparent through the hazard rate.